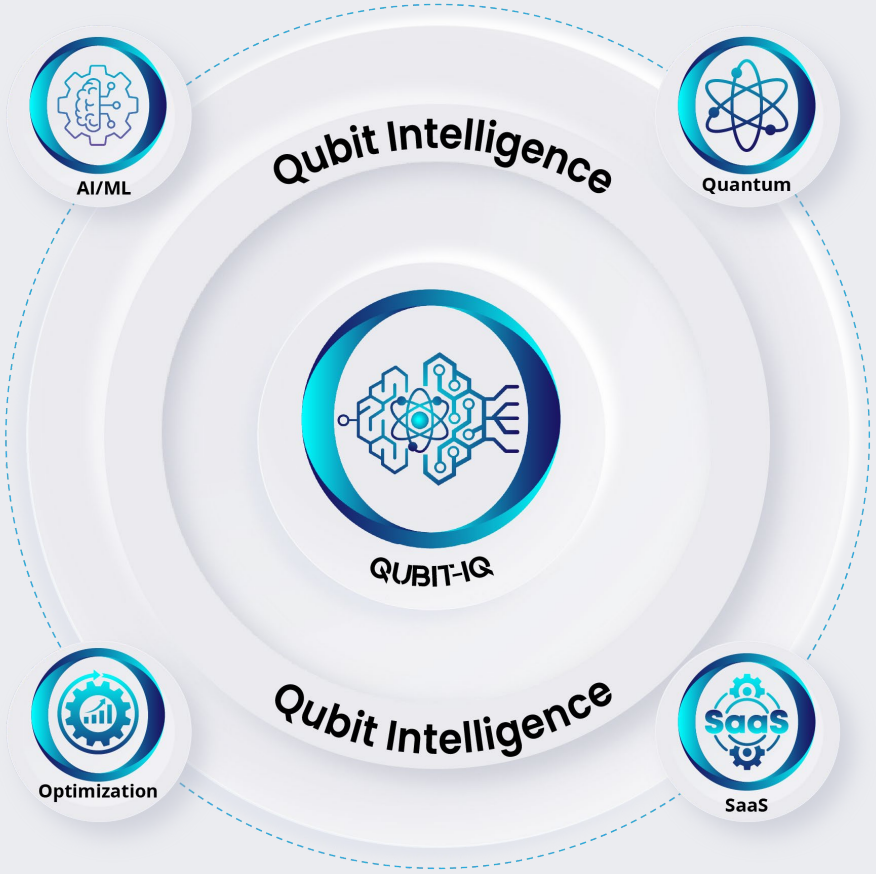
Leading the way in Quantum & AI Innovations

QUANTUM AI
GLOBAL

Qubit - Intelligence

Introduction



Qubit Intelligence: Generalized Quantum Algorithm Development Accelerator

"*Qubit-Intelligence*" built upon **"*Qubit-IQ*"** enables organizations with AI-driven Quantum Intelligence, accelerating optimal Quantum Algorithm development at scale.

Unified Platform:
Seamlessly integrates Classical, Quantum & Hybrid computing.

Intelligent Recommendations:
AI Augmented Encoding & Quantum Circuit Optimization to accelerate Quantum Algorithm development

Scalable & Problem-Agnostic:
Adapts to diverse domains and computational challenges.

Enterprise Acceleration:
Simplifies Quantum adoption across industries.

Real-World Impact:
Potential to drive breakthroughs in finance, logistics, energy, pharma, and manufacturing.

Qubit-Intelligence

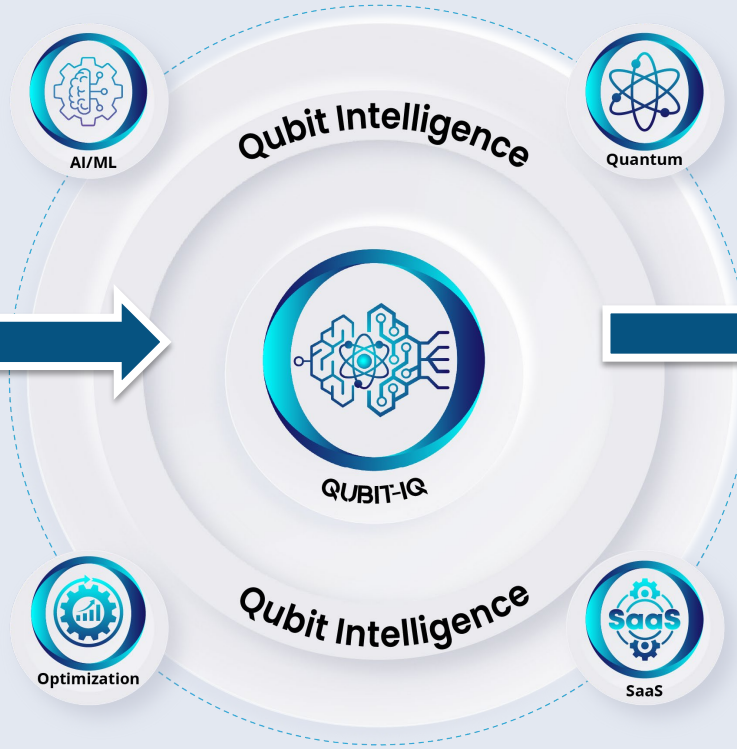
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PHASE - 1

Qubit-IQ

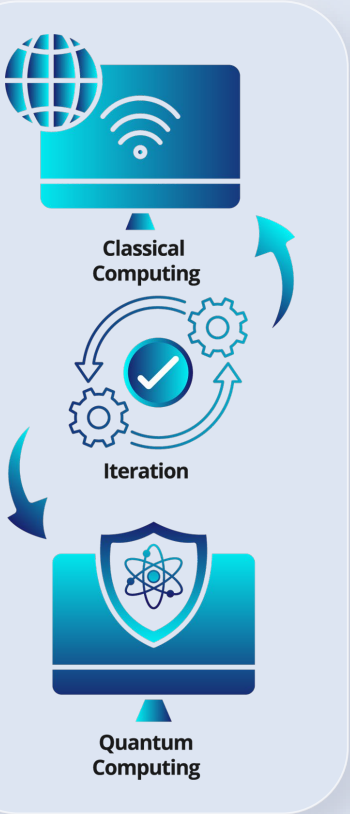
PHASE - 2

Qubit Intelligence



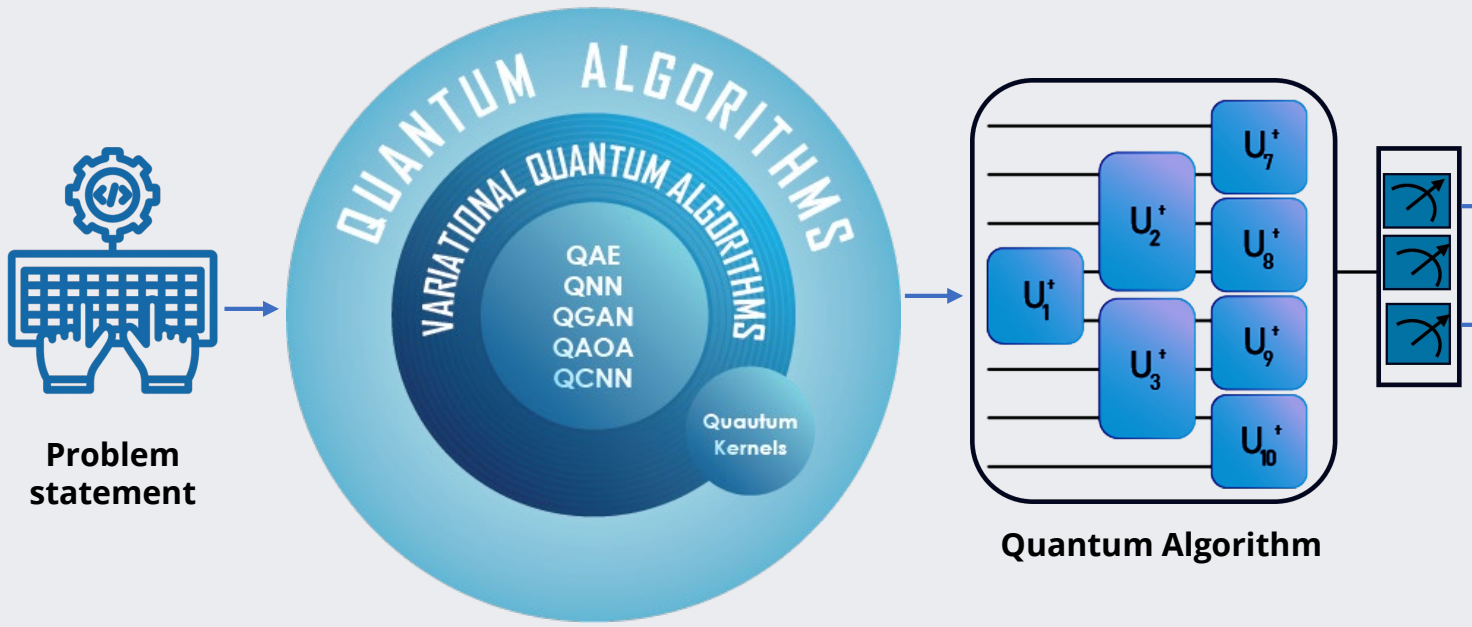
PHASE - 3

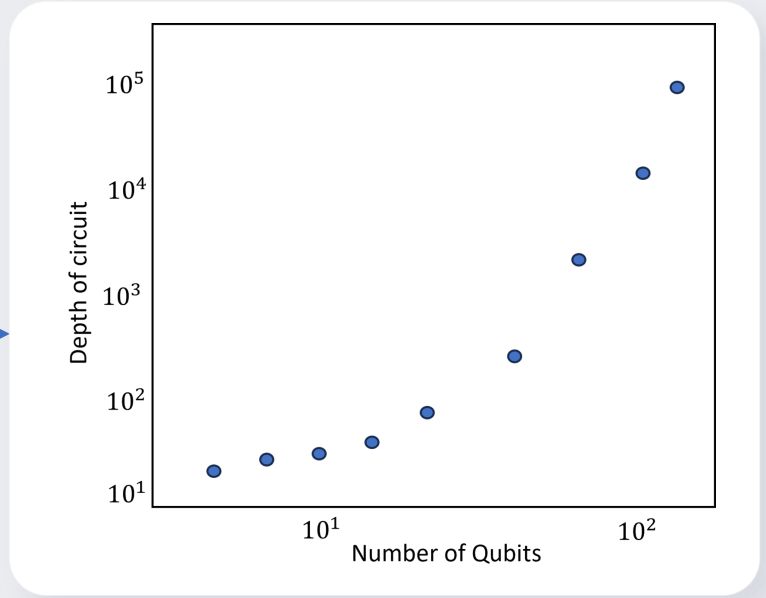
Generalized Quantum Algorithm Development Accelerator

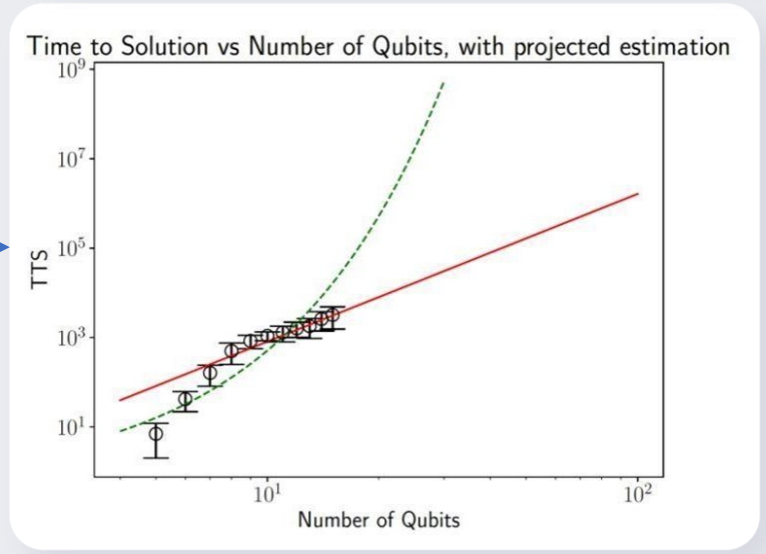


Problem statement

The surge in quantum algorithms complexity has made determining the suitable encodings and optimal circuits increasingly complex, slowing innovation and practical adoption.





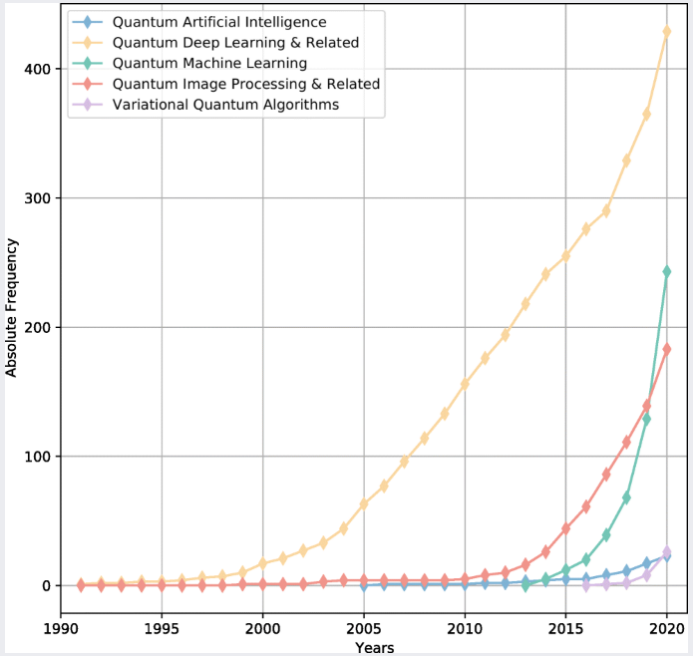


Pasqal, Towards quantum advantage with efficient graph implementations, 2022.

Market Opportunity

Evidence of research and data analysis supporting the problem and solution

- Research shows an exponential surge in quantum algorithms and PQCs, yet the lack of a standardized method for selecting optimal circuits remains a key challenge, hindering efficient problem-solving and innovation.



Quantum Algorithms	# of qubits	Pre-optimization		After Optimization		Improvement (%)	
		Execution time (S)	# of Gates	Execution time (S)	# of Gates	Execution time	Gates
VQC	3	6.4	21	6.3	18	1.56	14.28
Q transfer learning	4	6.8	28	6.2	25	8.82	10.71
Q neural network	10	6.9	67	6.6	57	4.34	14.92
Grover 1	4	6.4	41	6.2	21	3.12	48.78
Grover 2	4	6.7	39	5.9	27	11.94	30.76
TNN	4	6	27	5.9	14	1.7	48.14

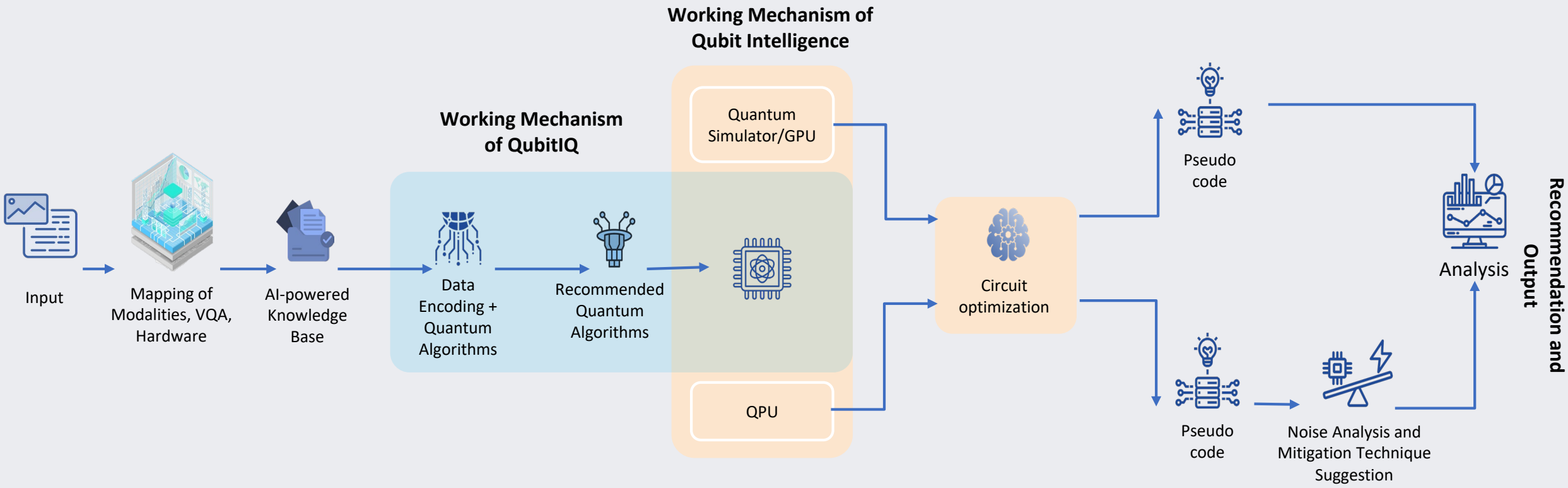
Specific Goals, Effectiveness and Impact

- **Precise:** Qubit Intelligence provides dedicated, context-aware quantum algorithm recommendations.
- **Productive:** Qubit Intelligence accelerates decision-making with AI-driven recommendations.
- **Efficient:** Qubit Intelligence reduces development time by 3-4x for users.
- **Use Qubit Intelligence:** Provide precise, context-aware recommendations, optimize search time and accelerate the development.
- **Use Generic LLMs:** Offers broader knowledge but lacks domain-specific precision, potentially increasing search time.

Flow Diagram

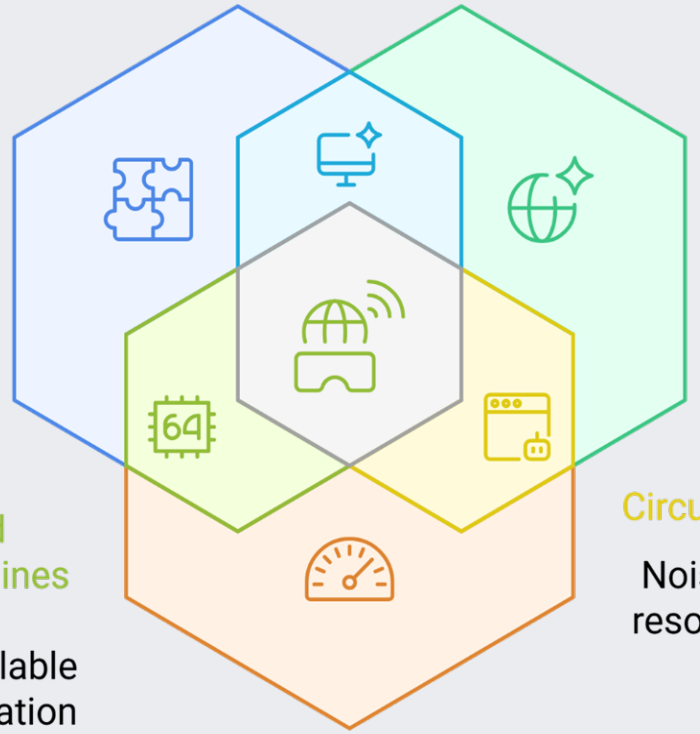
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Qubit Intelligence: Generalized Quantum Algorithm Development Accelerator



Innovation

Synergy Driving Quantum Innovation



Quantum-Specialized Knowledge Base

Deep understanding of quantum principles

AI-Driven Analytics

Intelligent data processing and insights

LLM-Integrated Benchmarking Engines

Real-time, scalable hardware evaluation

Circuit Optimization

Noise analysis and resource estimation

Quantum Algorithm Benchmarking

Evaluation of quantum algorithmic performance

Uniqueness

First AI-driven platform accelerating quantum algorithm development, VQA benchmarking, noise analysis, circuit optimization, and resource estimation within one adaptive ecosystem- offering precise, hardware-aware quantum insights.

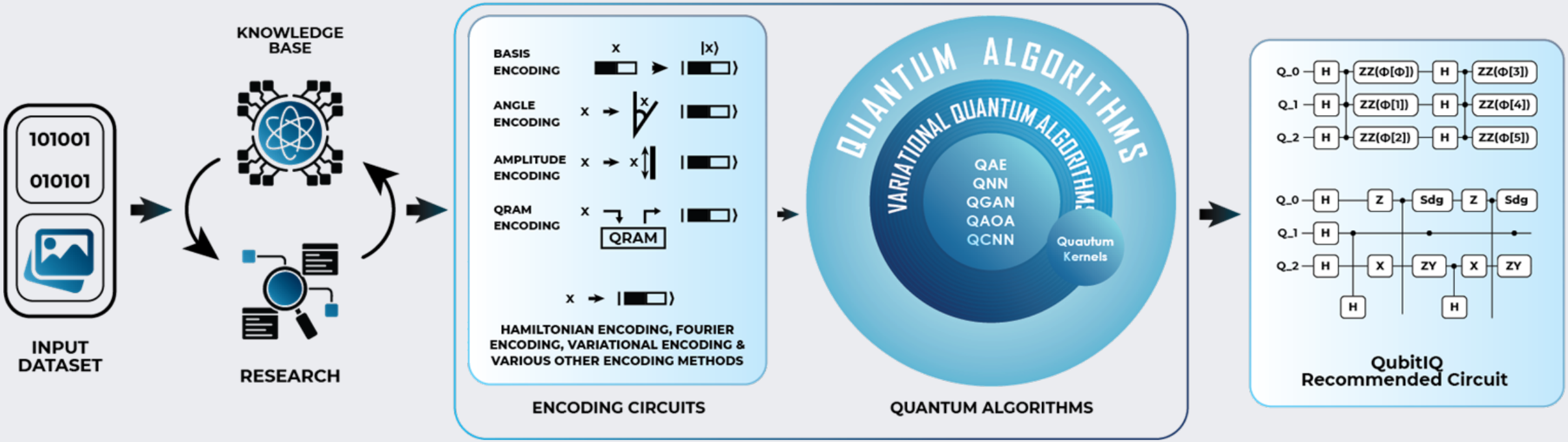
Key Differentiating Features

- **Accelerate Quantum Algorithm Development:** ML-guided encoding and circuit generation speed up algorithm design.
- **Real-Time Benchmarking**: Evaluates noise, optimization, and mapping dynamically.
- **Hardware Integration**: Scalable, resource-efficient optimization.
- **End-to-End Framework**: Beyond Qiskit or PennyLane - intelligent, unified, and enterprise-ready.

Comparison

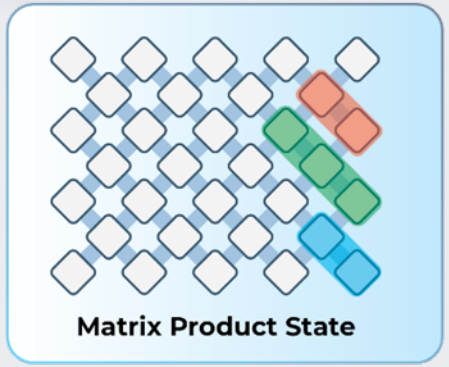
Features	Qubit Intelligence	IBM Qiskit	Xanadu PennyLane
ML Recommended Encoding	Problem statement that maps to the existing knowledge base for algorithms, and mapping	General-purpose SDK, no intelligent recommendation system	Limited to hybrid ML workflows, not domain-trained
VQA Optimization & Benchmarking	Integrated benchmarking across circuit depth, noise, and hardware scalability	Manual optimization required	Offers partial optimization tools, not comparative
Quantum Resource Estimation	Provides detailed estimation of qubits, gate depth, and hardware requirements	Not integrated	Minimal estimation tools

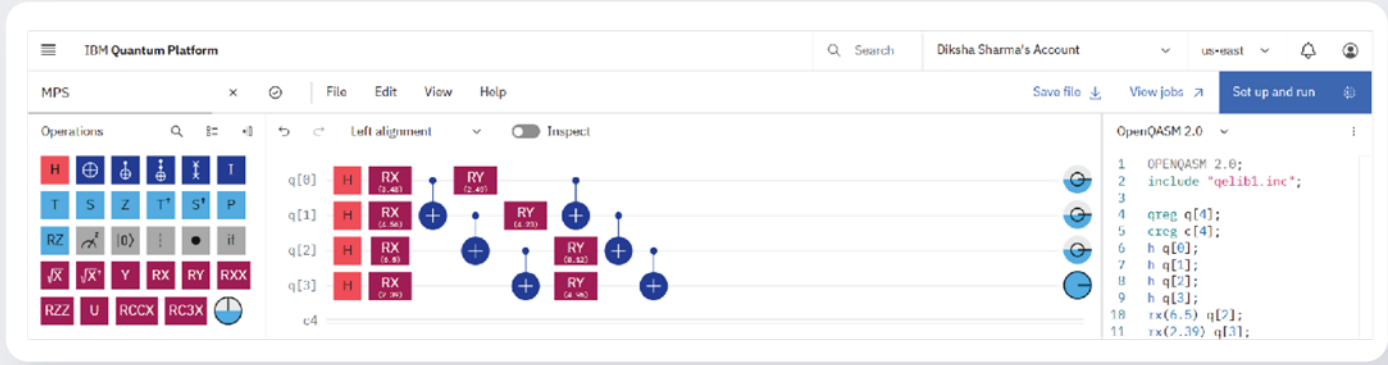
Demonstration



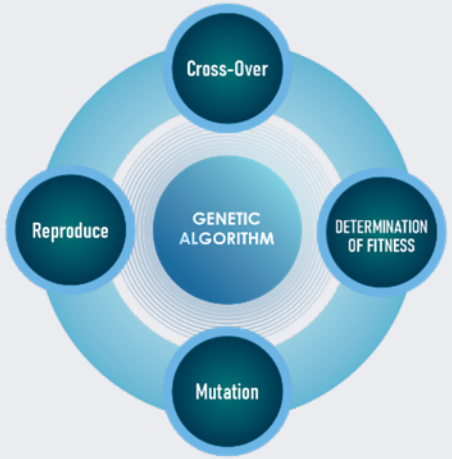
Demonstration

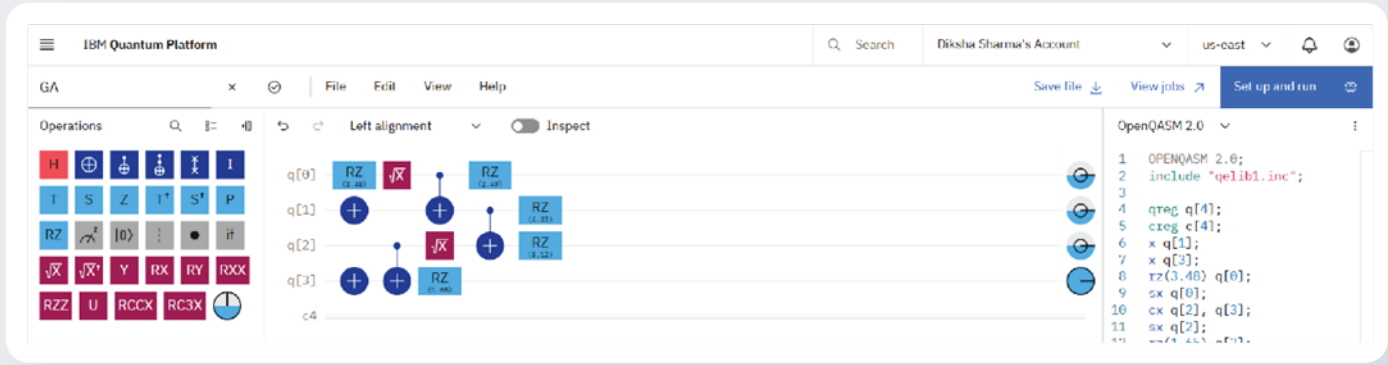
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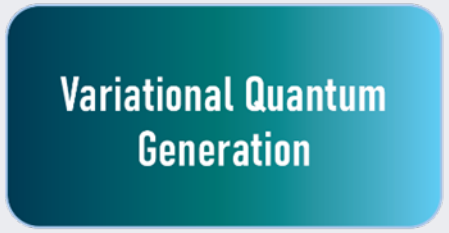


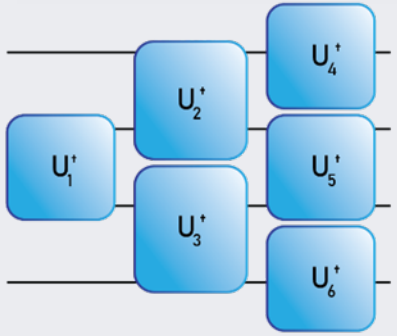


Qubit Intelligence Recommended Circuits



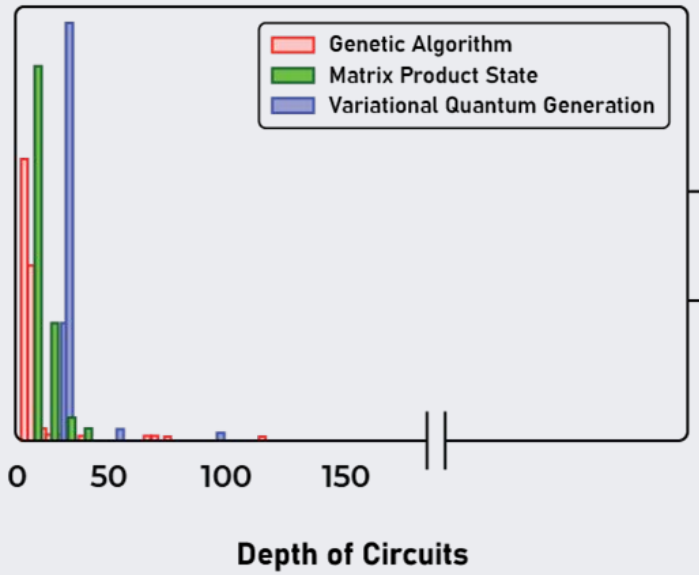




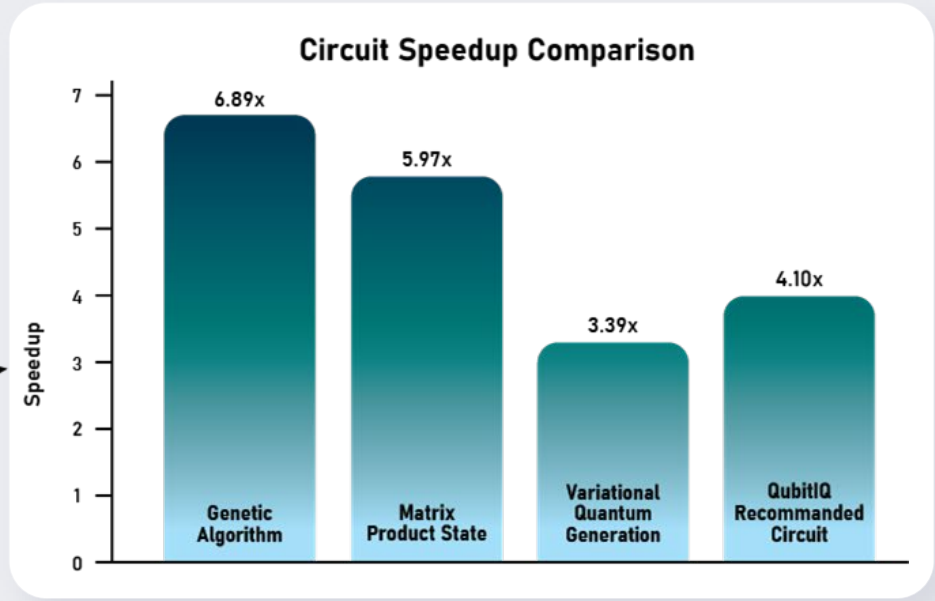


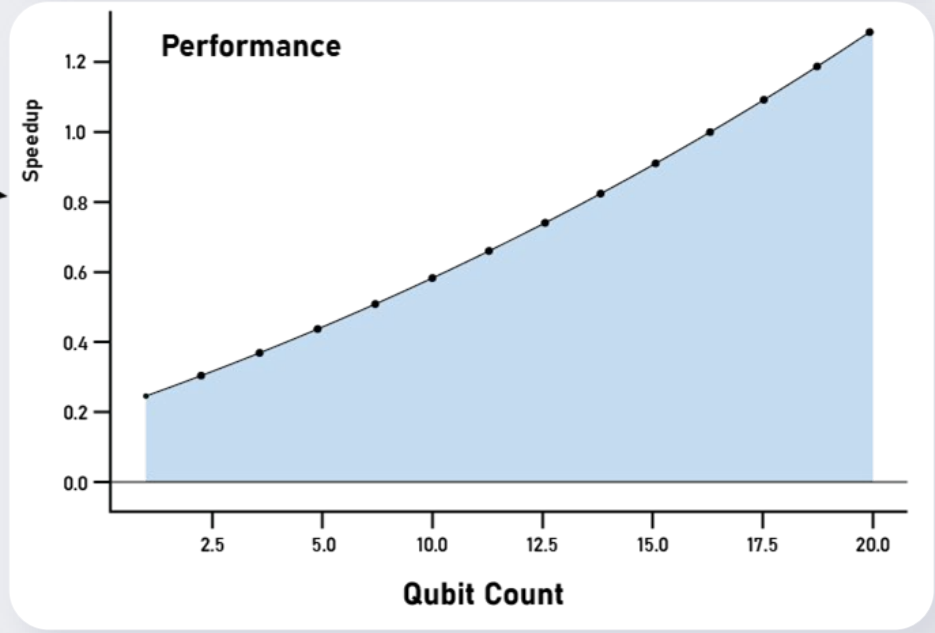
Composer | IBM Quantum Platform

Demonstration



Analysis





Composer | IBM Quantum Platform

Market Impact and Potential

Growth potential in the market or industry

Simulation	Optimization	Machine learning
($175 billion–$330 billion)	($100 billion–$220 billion)	($95 billion–$250 billion)
Pharma: Drug discovery $40 billion–$80 billion	Finance: Portfolio optimization $20 billion–$50 billion	Automotive: AV AI algorithms Up to $10 billion
Aerospace: CFD $10 billion–$20 billion	Insurance: Risk management $10 billion–$20 billion	Finance: AML and anti-fraud $20 billion–$30 billion
Chemistry: Catalyst design $20 billion–$50 billion	Logistics: Network optimization $50 billion–$100 billion	Tech: Search/advertising optimization $50 billion–$100 billion
Energy: Solar conversion $10 billion–$30 billion	Aerospace: Route optimization $20 billion–$50 billion	Other use cases $25 billion–$110 billion
Finance: Market simulation (e.g., derivates pricing) $20 billion–$35 billion		
Other use cases $75 billion–$115 billion		

https://www.bcg.com/publications/2024/long-term-forecast-for-quantum-computing-still-looks-bright

Target market size, potential revenue, and impact



https://dataintelo.com/report/quantum-middleware-platform-market

- The global quantum middleware market is expected to grow from US$856 million to US$7.6 billion.
- 29.7% CAGR boosts demand for quantum development tools, resulting in strong growth potential.

Boston Consulting Group, (2024), The Long-Term Forecast for Quantum Computing Still Looks Bright.
Data Intelo (2025), Quantum Middleware Platform Market Outlook.

Market Impact and Potential

01 **Potential for cost savings or increased efficiency**

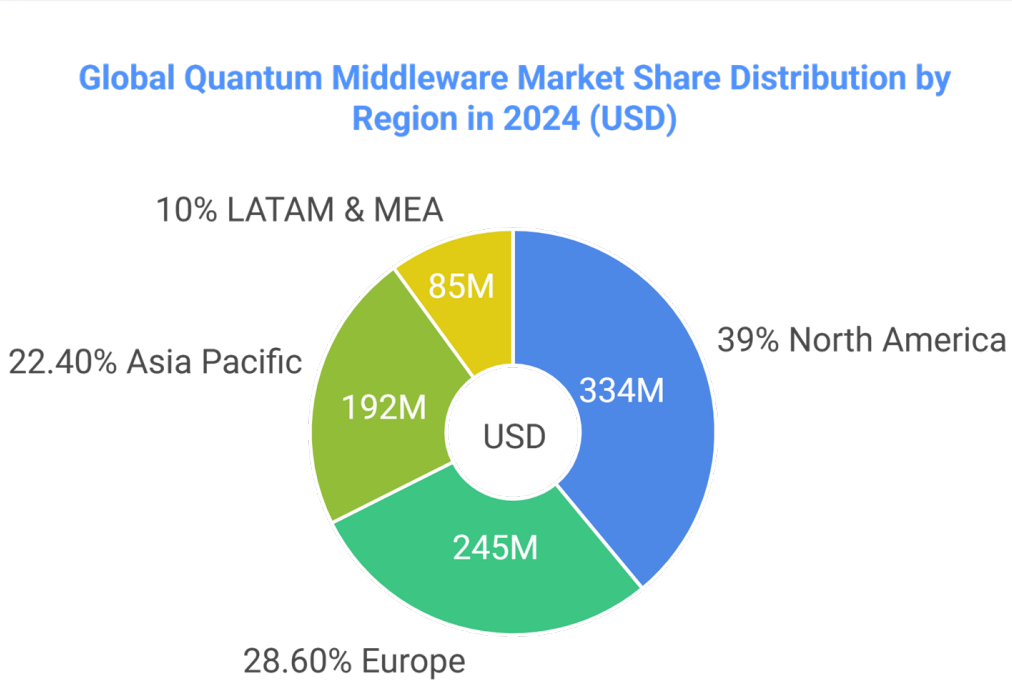
- Reduces integration and development costs through a unified middleware layer.
- Optimizes resource usage, lowering operational and infrastructure expenses.

02 **Target market segment, market share, and competitive landscape analysis**

- India’s quantum market to reach USD 231.8M by 2030 at 22.9% CAGR, with systems sector leading and services sector grow fastest.
- Growing adoption signals rising need for quantum middleware.

03 **Scalability and adaptability in different contexts, geographies, or industries**

- Domain-agnostic & scalable quantum middleware usable across pharma, manufacturing, finance, and logistics.
- Globally scalable, adapting to diverse industries and environments while maintaining consistent performance.



https://dataintelo.com/report/quantum-middleware-platform-market

Data Intelo (2025), Quantum Middleware Platform Market Outlook.
Grand View Research (2025), India Quantum Computing Market Size & Outlook, 2025-2030.

Social Impact and Sustainability

01 **Contribution to the betterment of society or the environment**

- Global data-centre demand rising from 350 GW (2022) to 1,000 TWh (2030).
- Qubit Intelligence delivers energy-efficient quantum-AI computing, reducing usage of high number of GPUs.

02 **Reduction of carbon footprint or environmental impact**

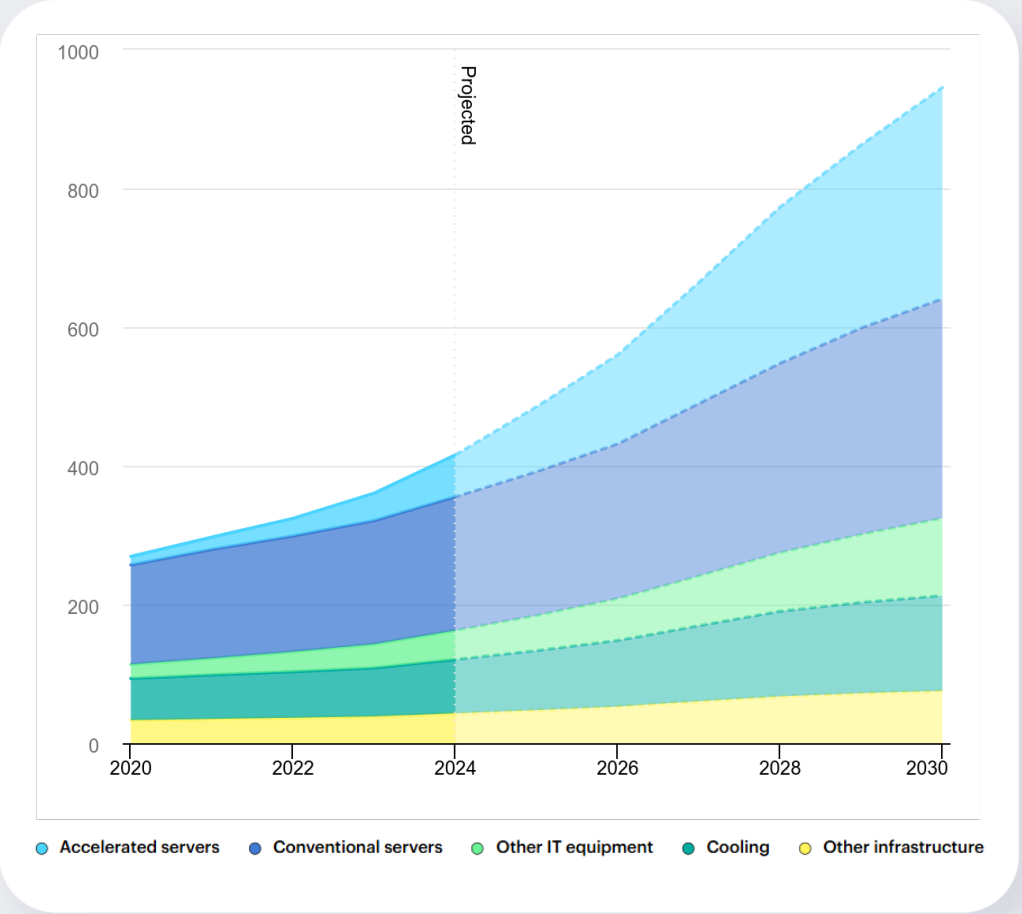
- Cuts GPU usage and overall power consumption.
- Supports sustainable, low-carbon digital workflows.

03 **Contribution to achievement of Sustainable Development Goals**

SDG 4, 8, 9, 13: Education, Growth, Innovation, Climate Action.

04 **Financial and environmental sustainability and profitability**

- Lowers cost and energy use through optimized simulations.
- Minimizes emissions by improving production and computational efficiency.



Global data centre electricity consumption, by equipment, Base Case, 2020-2030.

IEA, (2025), Energy demand from AI.

Conclusion

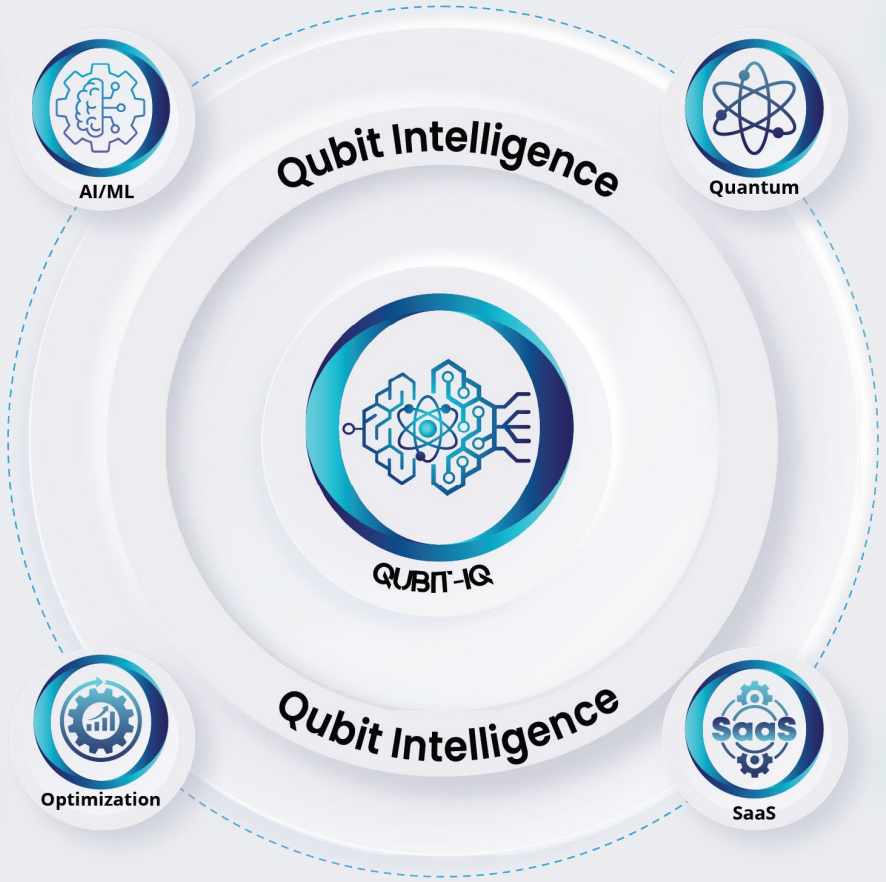
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1 **Accelerated Algorithm Development**

2 **Research to Innovation**

3 **Catalyst for Quantum Innovation**

4 **Problem-agnostic Design**



CORE TEAM

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Sanjay Chittore, Founder & CEO
Entrepreneur
deep understanding of the challenges and opportunities facing these industries



Col Kapil Jaiswal (Retd), CTO
Adjunct Prof, Cyber, Amrita VV, Ex-Director (InfoSec), MoD, GOI
Lead national level Info Security and Cryptography Research team of 100+ domain specialists and Cyber Operations team of 50+ Red Team members



Dr. Sachin Barthwal, Head of Quantum Technology
PhD in Physics from University of Hyderabad
Experimental Atomic Physicist with extensive experience of in Laser cooled Atomic Spectroscopy research including Quantum Sensor Technology



Dr. Diksha Sharma, Quantum Research Scientist
Research scholar in Quantum Machine Learning, IIT Jodhpur
Experienced in quantum machine learning models for real-world tasks, developing interpretable, scalable, and resource-efficient quantum models.



Dr. Monika, Quantum Sensing Lead
PhD in Physical Science from AcSIR–NPL
Experimental quantum optics physicist specializing in Rydberg atom based RF E-field sensing, quantum memory and atomic spectroscopy.



Dr. Avinash Kumar, Quantum Optics Scientist
PhD in Physics , Tata Institute of Fundamental Research
Experimental physicist specialized in designing and building narrow linewidth laser cavities with expertise in optics and high-resolution spectroscopy.



Dr. Dhanshre Thulkar, Head of Product Management
PhD in Engineering from University of Mumbai
Researcher in machine learning and image processing and experience in working with industry and academia



Diana Dsouza, Head of Sales and Business Development
Business Development leader with 18 years expertise in end-to-end product commercialization. Adept at ensuring successful market entry and driving robust sales performance.



Rikteem Bhowmick, Quantum Communications Lead
Research scholar pursuing PhD in Quantum Communication
Experienced in developing entangled photon sources and designing quantum key distribution systems.



Devendra Mishra, Quantum Hardware Lead
Research scholar pursuing PhD in Quantum Optics IITH
Experienced in Quantum Optics experimentation and Quantum communication Hardware (Quantum memory, Quantum Repeater, Etc.)



Rama Theja, Opto Electronics Lead
Masters in Quantum computing from DIAT Pune
Quantum engineer with expertise in setting up quantum sources, QKD testbeds, and quantum memory hardware with integrated control electronics.



Neelesh Singh Katoch, Quantum Software Engineer
Masters in Cybersecurity from DIAT Pune
Experienced engineer and researcher working on quantum-safe cryptography and cybersecurity, designing and implementing scalable, standards-compliant production solutions.